Working with MIMIC-IV

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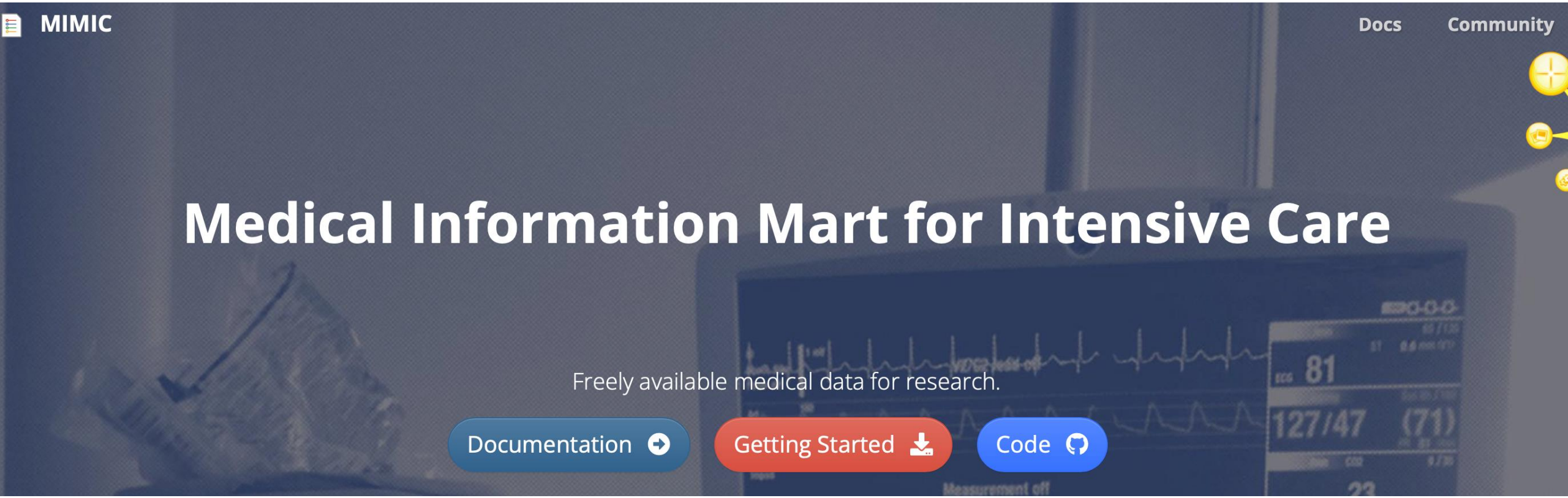
Outline

- MIMIC-IV
- Exercise of Assignment 1
- More Material


MIMIC
Docs
Community
Medical Information Mart for Intensive Care
Freely available medical data for research.
Documentation
Getting Started
Code

MIMIC-IV

MIMIC can help understand

- ICU prediction tasks
- Temporal data structure
- Data leakage issues

Tasks handled with MIMIC

Clinical problems including:

- Predicting in-hospital mortality
- Detecting physiologic decline
- Phenotype classification
- Forecasting length of stay

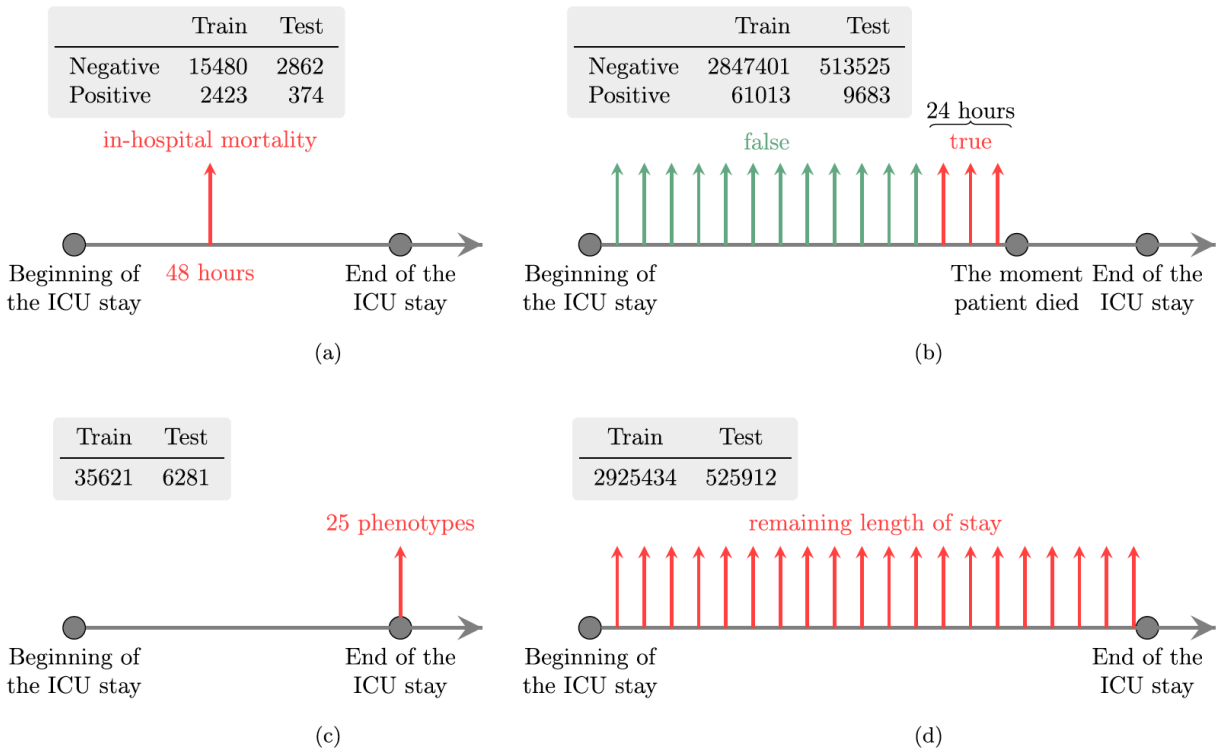


Fig. 1 Summaries of the four benchmark tasks. Each subfigure consists of two parts. The table lists number of prediction instances for the corresponding task. The timeline shows when the predictions are done. Note that in the decompensation and length-of-stay prediction tasks predictions are done hourly, and each vertical arrow corresponds to one prediction instance. (**a**) In-hospital mortality. (**b**) Decompensation. (**c**) Phenotyping. (**d**) Length of stay.

Harutyunyan, H., Khachatrian, H., Kale, D.C. *et al.* **Multitask learning and benchmarking with clinical time series data**. *Sci Data* **6,** 96 (2019). https://doi.org/10.1038/s41597-019-0103-9

Assignment 1

Objective

- Prediction of the length of stay (LOS) of the ICU patients ("short" or "long") based on the **severity score OASIS** computed by using measures of the first 24 hours (and 48 hours) of an ICU stay on MIMIC-IV.

- Using Demo dataset:
  - The demo dataset contains all intensive care unit (ICU) stays for 100 patients (randomly selected from a subset of patients).

Severity of illness scores

- **APACHE IV:** The **A**cute **P**hysiology, **A**ge, and **C**hronic **H**ealth **E**valuation system provides two assessments of patient health:
  - Acute Physiology Score (**APS-III**) derived from 17 physiologic parameters.
  - APACHE probability of mortality
- **SAPS III:** Simplified Acute Physiology Score
- **OASIS:** Oxford Acute Severity of Illness Score

- Mostly used for prediction of:
  - ICU and in-hospital mortality
  - Length of stay

Severity of illness scores

- The OASIS uses: age, Pre-ICU LOS, elective surgery and the following 7 physiologic measurements (worst value of the first 24h):
  - Glasgow Coma Score (GCS)
  - Heart rate
  - Mean arterial pressure (MAP)
  - Respiratory rate
  - Temperature
  - Urine output
  - Ventilated

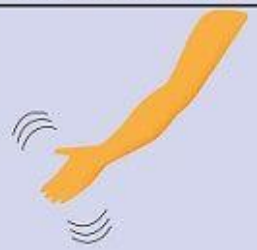
OASIS is the sum of the following values:

5 <0.17	**3** 0.17-4.94		Pre-ICU LOS **0** 4.95-24.00 Hours	**2** 24.01-311.80	**1** >311.80			
			Age **0** <24 Years	**3** 24-53	**6** 54-77	**9** 78-89	**7** >90	
10 3 - 7	**4** 8 - 13	**3** 14	GCS **0** 15					
		4 <33	Heart Rate **0** 33-88 min^{-1}	**1** 89-106	**3** 107-125	**6** >125		
4 <20.65	**3** 20.65-50.99	**2** 51-61.32	MAP **0** 61.33-143.44 mmHg	**3** >143.44				
	10 <6	**1** 6-12	Respiratory Rate **0** 13-22 min^{-1}	**1** 23-30	**6** 31-44	**9** >44		
3 <33.22	**4** 33.22-35.93	**2** 35.94-36.39	Temperature **0** 36.40-36.88 °C	**2** 36.89-39.88	**6** >39.88			
10 <671	**5** 671-1426.99	**1** 1427-2543.99	Urine Output **0** 2544-6896 Cc/day	**8** >6896				
			Ventilated **0** NO	**9** YES				
		6 NO	Elective Surgery **0** YES					

https://alistairewj.github.io/project/oasis/

Glasgow Coma Scale

Medical scale designed to assess the depth (and duration) of impaired consciousness and coma states.

Behaviour	Response
Eye Opening Response	4. Spontaneously 3. To speech 2. To pain 1. No response
Verbal Response	5. Oriented to time, person and place 4. Confused 3. Inappropriate words 2. Incomprehensible sounds 1. No response
 Motor Response	6. Obeys command 5. Moves to localised pain 4. Flex to withdraw from pain 3. Abnormal flexion 2. Abnormal extension 1. No response

Source: https://biologydictionary.net/glasgow-coma-scale/

More Details of Assignment 1

- Predict Length of Stay (LOS) in ICU.
- Transform the problem to a **binary classification task** using the binarized LOS:
  - LOS < 4 days → short stay
  - LOS ≥ 4 days → long stay
- Use OASIS score.
- Other alternatives.

- Download the notebook from Campus Virtual.

More Details of Assignment 1 (II)

- Before compute OASIS:
- Normalize times from the time the patient was transferred into the ICU to identify the measurements of the first 24 hours (or 48 hours) as observations.
- Implement different strategies:
  - All the instances of the first 24 hours.
  - All the instances of the first 48 hours.
- MIMIC has many NaN values.
  - The original OASIS score is NaN if any of the considered variables has all its instances as NaN (https://gist.github.com/tompollard/d4c216d0c4ad30481c99)
  - Alternative: consider the variable as 0.0 (state in normal range) when all instances of the variable are NaN.

More Material

Tutorials of MIMIC-IV

Videos:

1. https://alistairewj.github.io/talk/2020-mimic-iv-data-tutorial/
2. https://alistairewj.github.io/talk/2020-mimic-iv-analysis-tutorial/
   - Code of tutorial 2: https://github.com/alistairewj/mimic-iv-aline-study/blob/main/mimic_iv_aline.ipynb

- Tutorial MIMIC-IV: https://colab.research.google.com/drive/1REu-ofzNzqsTT1cxLHIegPB0nGmwKaM0?usp=sharing

References

For more details about benefits of EHR adoption:

- Casey et al. *Using Electronic Health Records for Population Health Research: A Review of Methods and Applications*. Annu Rev Public Health. 2016;37:61-81.doi:10.1146/annurev-publhealth-032315-021353
- Campanella et al. *The impact of electronic health records on healthcare quality: a systematic review and meta-analysis*, European Journal of Public Health, Volume 26, Issue 1, February 2016, Pages 60–64, https://doi.org/10.1093/eurpub/ckv122
- Menachemi N, Collum TH. *Benefits and drawbacks of electronic health record systems*. Risk Manag Healthc Policy. 2011;4:47-55. doi:10.2147/RMHP.S12985
- Bertelsmann Report (https://www.bertelsmann-stiftung.de/fileadmin/files/Projekte/Der_digitale_Patient/VV_SHS-Studie_EN.pdf)

References

- Benjamin Shickel, Patrick Tighe, Azra Bihorac, Parisa Rashidi. Deep EHR: A Survey of Recent Advances in Deep Learning Techniques for Electronic Health Record (EHR) Analysis. https://arxiv.org/abs/1706.03446
- Johnson, A., Pollard, T., Shen, L. *et al.* MIMIC-III, a freely accessible critical care database. *Sci Data* **3,** 160035 (2016). https://doi.org/10.1038/sdata.2016.35

To complement today's class

- Why electronic health records? https://www.youtube.com/watch?v=Lo_3qOejQzI&feature=emb_imp_woyt
- Electronic Health Records: Usability and Unintended Safety Issues: https://www.youtube.com/watch?v=SlQ60KaLeHg
- Interoperability of Electronic Health Records– Benefits and Opportunities: https://www.youtube.com/watch?v=gWfzcOliRac
- European Health Information Gateway: https://gateway.euro.who.int/en/search/?q=EHR#page_indicators=1
- FHIR: https://www.youtube.com/watch?v=OIt0GrCPu8k

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Data Science for Health